AeroGraph: Hybrid Graph–Vector Retrieval with Dual-Judge Evaluation over Aviation Safety Narratives

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Abstract

Flat dense retrieval underperforms on multi-hop causal questions over aviation safety incident reports: relevant evidence is spread across many short narratives that share only a chain of entities. We present AeroGraph, a system that constructs a directed knowledge graph from 2,000 real NASA Aviation Safety Reporting System (ASRS) reports via ontology-constrained LLM extraction, canonicalizes entities against HFACS, ICAO, weather, and flight-phase taxonomies, and fuses four retrieval signals (dense vector, sparse BM25, Personalized PageRank, and Leiden-community summaries) through Reciprocal Rank Fusion. Raw extraction yields 29,244 entities and 43,505 relations; canonicalization and synonym expansion reduce this to 23,948 entities and 48,479 relations across 186 weakly connected components. We evaluate six retrieval configurations on a 50-query benchmark with paired Wilcoxon significance tests and two independent judges: a local `llama3.1:8b` and Claude Sonnet. The 4-signal hybrid significantly improves precision@10 over a vector-only baseline (0.210 vs. 0.114, $p = 0.047$), as does vector-plus-2-hop-graph (0.152, $p = 0.018$). Personalized PageRank alone reaches P@10 = 0.200 but is statistically indistinguishable from both the vector baseline ($p = 0.16$) and the full hybrid ($p = 0.23$); its apparent lead is within the noise floor at $n = 50$. Under blind pairwise preference all three graph-augmented systems significantly beat the vector baseline (GraphRAG 72%, $p = 0.018$; Hybrid 62%, $p = 0.047$; PPR wins the decisive-query subset against GraphRAG at $p = 0.036$), but are statistically interchangeable with one another (Hybrid vs. PPR $p = 0.081$; Hybrid vs. GraphRAG $p = 0.50$). Pointwise Claude-rubric faithfulness and pairwise Claude preference disagree on the same answers; we argue these two protocols measure different constructs and future RAG benchmarks should report both. Rankings of the bottom three systems are preserved across judges; top-three rankings disagree by judge. Code, taxonomy tables, benchmark queries, and per-query judge scores are released at https://github.com/Aryan95614/AeroGraph.

1 Introduction

The U.S. Federal Aviation Administration holds more than 15,000 public near-miss reports annually, a figure that has roughly doubled over the past decade. Each report is a short, unstructured narrative written by a pilot, controller, or maintenance technician describing an incident and its contributing factors. Individually, reports are low value; they describe events that almost happened. Collectively, they encode the operational precursors of serious accidents. The January 2025 mid-air collision near Reagan National Airport, which killed 67 people, involved contributing factors (high-density traffic, altitude-clearance ambiguity, visual-approach workload) that appear repeatedly in ASRS narratives filed over the preceding decade.

Surfacing those patterns at scale requires retrieval that is both recall-sensitive (finding the handful of relevant reports among many) and capable of multi-hop reasoning (combining evidence from reports that share no vocabulary but do share a chain of entities). Flat dense retrieval, the standard baseline, fails on both counts. It returns surface-form similar passages while missing semantically related reports that describe the same causal chain with different terminology, and it provides no mechanism for following an entity through a graph of co-occurrences.

This paper presents AeroGraph, a hybrid retrieval system for aviation safety reports, and a careful benchmark of six retrieval configurations. We focus on three questions. (i) Does graph augmentation improve retrieval and answer quality over a vector baseline? (ii) Which signal or combination of signals drives that improvement? (iii) How robust are the conclusions to the choice of LLM judge?

Contributions.

- A working pipeline combining dense vector retrieval, BM25, Personalized PageRank [7], and Leiden-community summaries [4, 14] over a domain knowledge graph built from 2,000 ASRS reports.
- A 50-query benchmark (single-hop, multi-hop, comparative) and six-system ablation, with paired Wilcoxon signed-rank significance tests and 95% confidence intervals reported per metric. Benchmark queries, per-query outputs, and per-query judge scores are released.
- A dual-judge protocol (local `llama3.1:8b` and Claude Sonnet) that exposes a 0.2 to 0.4 absolute-faithfulness gap on identical outputs. We show that pointwise Claude-rubric faithfulness and pairwise Claude preference disagree on the same answers and argue they measure different constructs.
- An empirical finding with clear implications for RAG evaluation: Personalized PageRank alone attains P@10 statistically indistinguishable from a 4-signal hybrid ($p = 0.23$) but loses pairwise preference under a blind judge. Retrieval precision at a fixed cutoff does not track answer quality, and a higher-top-rank retriever is not necessarily a better one.

2 Related Work

Retrieval-augmented generation. Lewis et al. [10] established the retrieve-then-read pattern. Later work extends the basic architecture. Fusion-in-Decoder [8] encodes passages independently before decoder fusion. RETRO [2] scales retrieval to trillion-token datastores. HyDE [6] retrieves against a hypothetical answer. Self-RAG [1] trains a model to decide when and what to retrieve. A persistent limitation is that flat retrieval over independent chunks struggles with multi-hop reasoning [15]: the evidence needed is distributed across documents that do not individually match the query.

Graph-based retrieval and GraphRAG. Knowledge-graph augmentation of language models predates modern RAG. Sun et al. [13] and Yasunaga et al. [18] integrate structured knowledge-base traversal with language-model reasoning for commonsense QA. The term *GraphRAG* was established by Edge et al. [4], whose two-stage pipeline (LLM extraction followed by Leiden community detection [14] and hierarchical summarization) yields substantial gains on global sense-making queries over news and podcast corpora. Gutiérrez et al. [7] propose HippoRAG, which replaces community summaries with Personalized PageRank over an entity graph and attains strong multi-hop QA performance. Our hybrid retriever follows HippoRAG in using PPR as a core signal, follows GraphRAG in using community summaries for global queries, and adds dense vector and sparse BM25 as complementary signals within a single Reciprocal Rank Fusion [3] pipeline.

NLP for aviation safety. ASRS has been studied primarily through text classification and topic modeling [9, 11]. Formal taxonomies such as HFACS [12] and ICAO occurrence categories [**?**] provide canonical vocabulary for human and technical contributing factors. We use HFACS Level-2 categories and ICAO aircraft and airport codes as canonical lookup targets during entity resolution, a practice standard in aviation safety analytics but, to our knowledge, not previously applied inside an LLM-based GraphRAG pipeline.

LLM-based information extraction. LLMs attain competitive zero-shot NER performance [17] but suffer on fine-grained entity distinctions and introduce fragmentation: the same concept is extracted as multiple distinct nodes under slight surface-form variation. Wadhwa et al. [16] document the same tradeoff for relation extraction. We mitigate fragmentation with ontology-constrained prompting, number-aware fuzzy deduplication, and a three-tier taxonomy resolution cascade (exact match; `rapidfuzz` edit-distance at threshold 85; embedding cosine similarity at 0.75).

RAG evaluation. RAGAS [5] decomposes RAG evaluation into faithfulness, answer relevance, context precision, and context recall. LLM-as-judge is now standard [19], with documented position, verbosity, and self-enhancement biases that affect absolute but rarely relative scores. We adopt RAGAS-style metrics and mitigate self-enhancement bias by evaluating under two independent judges: one local open-weight model and one frontier API model.

3 Dataset and Knowledge Graph

3.1 ASRS Corpus

We sample 2,000 reports from the public `elihoole/asrs-aviation-reports` HuggingFace dataset (38,655 total records). Each report combines two narrative fields (Reporter 1, Reporter 2) with structured metadata: ACN (accession number), aircraft make and model, phase of flight, and anomaly classification. Aircraft strings are normalized to canonical ICAO designators at ingestion, so *Boeing 737-800*, *B737-800*, and *737* all map to B737. Float-formatted ACNs are normalized to integers. Reports with fewer than fifty narrative words are filtered.

3.2 Ontology

The aviation safety ontology has 10 node types (Aircraft, Event, Phase, Factor, Component, Outcome, Recommendation, ATC_Facility, Weather, TimePeriod) and 8 edge types (CAUSED_BY, CONTRIBUTED_TO, OCCURRED_DURING, INVOLVED, RESOLVED_BY, PRECEDED_BY, CO_OCCURRED_WITH, TEMPORAL_SEQUENCE). The ontology is intentionally coarser than formal safety taxonomies such as ECCAIRS and CAST, since richer types exceed reliable LLM extraction at our corpus size.

3.3 Taxonomy Tables

Normalization uses five canonical lookup tables totaling 1,157 surface-form aliases: HFACS Level-2 (19 categories, 152 aliases), ICAO aircraft designators (37 types, 215 aliases), airports (154 ICAO codes, 571 aliases), weather conditions (20 categories, 119 aliases), and flight phases (12 phases, 100 aliases). All lookup data ships with the public repository.

3.4 Extraction

Reports are processed in batches of 20 by Claude Sonnet with an ontology-constrained structured-JSON prompt. The system message specifies the node and edge types; the user message contains one report's narrative and metadata. Retry logic with exponential backoff handles API rate limits. Results are cached; re-runs are idempotent.

Post-extraction normalization has three stages. **(1) Name canonicalization** applies lowercasing, whitespace collapse, trailing-punctuation removal, and aviation abbreviation expansion. **(2) Fuzzy deduplication** merges within-type names with `SequenceMatcher` similarity above 0.85; a number-aware guard prevents merges of entities that differ only by a numeric identifier, so *engine 1 failure* $\neq$ *engine 2 failure* and *runway 28L* $\neq$ *runway 28R*. **(3) Taxonomy resolution** applies a three-tier cascade: exact match against canonical aliases, then `rapidfuzz` edit-distance at threshold 85, then sentence-transformer embedding cosine similarity at threshold 0.75.

3.5 Graph Construction

Extracted entities and relations merge into a directed graph with Merge semantics: nodes key on (type, canonical_name); edges on (source, target, type). Duplicate edges increment a weight counter and accumulate report-ID lists. Two backends are supported (Neo4j for production, NetworkX as in-process fallback) exposing a common query interface: `get_neighbors`, `get_causal_chain`, `get_subgraph`, `get_temporal_chain`. Two-hop BFS expansion skips nodes with degree > 200, so high-frequency hub entities do not dominate neighborhoods.

Graph statistics. Raw extraction produces 29,244 entities and 43,505 relations but 3,766 weakly connected components, of which 3,367 are isolated nodes, a well-documented failure mode of LLM extraction. Taxonomy normalization collapses entities to 23,948 canonical nodes with 48,479 relations and reduces the component count to 245. A synonym-edge pass (adding SYNONYM edges between canonical-alias pairs) reduces further to 186 components. Leiden clustering over the canonicalized graph yields 311, 393, and 296 communities at three resolution levels; we precompute LLM summaries for the 100 largest communities.

4 Hybrid Retrieval

Four signals are computed in parallel, then fused.

Dense vector. Reports are chunked into 256-token windows with 128-token overlap and embedded with `sentence-transformers/all-MiniLM-L6-v2` [?] (384 dimensions). Chunks are stored in ChromaDB (4,710 chunks). The top $2k$ chunks are retrieved by cosine similarity at query time.

Sparse BM25. A from-scratch Okapi BM25 implementation [?] with $k_1 = 1.5$ and $b = 0.75$ indexes the same chunk corpus. The inverted index is built at query time from the ChromaDB corpus with no external library.

Personalized PageRank. Query entities are extracted via keyword match against the entity index (with Claude fallback) and serve as the personalization vector for a PageRank computation [?] with

$\alpha = 0.15$. Nodes are ranked by stationary probability; reports are scored by the summed probability of their constituent entities.

Community summaries. For queries classified as global (via a lightweight keyword heuristic), the 100 community summaries are scored by BM25 against the query. The top 10 summaries pass as additional context.

Reciprocal Rank Fusion. Signals are fused with RRF [3]:

$$\mathrm{RRF}(d) = \sum_{s \in \mathcal{S}} \frac{w_s}{k + \mathrm{rank}_s(d)}, \tag{1}$$

where $k = 45$ (tuned from the standard 60 on a held-out validation split) and $\mathcal{S} = \{\text{vector}, \text{BM25}, \text{PPR}, \text{community}\}$. Signal weights are $w_{\text{vector}} = 1.0$, $w_{\text{BM25}} = 1.0$, $w_{\text{PPR}} = 1.5$, $w_{\text{community}} = 1.3$, tuned on the validation subset.

Answer generation. The top-k chunks, PPR-highlighted entities, and applicable community summaries pass to Claude Sonnet with a system prompt establishing the role of an aviation safety analyst. The prompt instructs the model to cite specific ACN identifiers, mark unsupported claims as such, and use causal language only when supported by CAUSED_BY or CONTRIBUTED_TO edges.

5 Experimental Setup

5.1 Benchmark

The 50-query benchmark has three query types. *Single-hop* factual queries ($n = 20$) test fact retrieval from one report (for example: *What weather condition is most commonly associated with go-around decisions?*). *Multi-hop* causal queries ($n = 20$) test reasoning across reports (*What causal chain links bird strikes to engine failure and subsequent go-around decisions?*). *Comparative* queries ($n = 10$) test cross-corpus comparison (*Compare contributing factors in B737 vs. A320 engine failure incidents*). Queries were constructed by the author to exercise the ontology and are released with the benchmark.

5.2 Retrieval Systems

1. **Vector**: ChromaDB dense-only retrieval.
2. **BM25**: sparse-only retrieval over the chunk corpus.
3. **Graph-only**: 2-hop BFS neighborhood expansion from query entities; reports scored by matched-entity coverage.
4. **GraphRAG**: vector and 2-hop graph fused via RRF, as in Edge et al. [4] but without community summaries.
5. **PPR-only**: Personalized PageRank on the entity graph, HippoRAG's core retrieval signal [7].
6. **Hybrid RRF**: 4-signal fusion (vector + BM25 + PPR + community summaries).

All systems share the same generation backend (Claude Sonnet with identical system prompt), isolating retrieval as the experimental variable.

5.3 Metrics and Significance

Retrieval. Precision@10, Recall@10, and nDCG@10 against graph-derived relevance labels. A report r is relevant to query q if q's extracted entities intersect r's entity set. Entities with degree > 200 are excluded from the relevance computation to prevent trivial matches on hub nodes. We discuss the circularity of this oracle in Section 8.

Generation. Faithfulness and answer relevance are scored by two independent judges. The *Ollama judge* is a local `llama3.1:8b` model with a four-criterion binary rubric: entity grounding, causal grounding, claim support, and no fabrication. The *Claude judge* is Claude Sonnet with the same rubric on a $\{0, 0.25, 0.5, 0.75, 1\}$ scale. Causal chain accuracy is scored only on multi-hop queries and requires both a present causal chain and a coherent ordering.

Pairwise preference. For each query and each pair of systems, both answers are submitted to the Claude judge with system labels blinded and positions randomized, producing a preference verdict (win, lose, tie).

Table 1: Retrieval quality on the 50-query benchmark. Mean $\pm$ std. Δ is the mean per-query difference against the vector baseline, with paired Wilcoxon signed-rank p-value (one-sided, system > baseline). Best value per column in **bold**; p-values < 0.05 are marked $*$.

System	P@10	nDCG@10	Latency (s)	Δ P@10	p
Vector baseline	0.114 ± 0.190	0.121 ± 0.209	1.6	—	—
BM25	0.106 ± 0.166	0.116 ± 0.192	**0.7**	-0.008	0.53
Graph-only	0.058 ± 0.169	0.072 ± 0.191	**0.7**	-0.056	0.92
GraphRAG	0.152 ± 0.231	0.151 ± 0.237	19.2	$+0.038$	$\mathbf{0.018}^{*}$
PPR-only	0.200 ± 0.349	0.221 ± 0.397	2.6	$+0.086$	0.16
Hybrid RRF	$\mathbf{0.210 \pm 0.301}$	$\mathbf{0.231 \pm 0.361}$	23.1	$+0.096$	$\mathbf{0.047}^{*}$

Table 2: Dual-judge answer quality. Ollama (`llama3.1:8b`, local) and Claude (Claude Sonnet) score independently. Causal accuracy is scored only on multi-hop queries ($n = 20$). Means across all 50 queries. Best per column in **bold**.

	Ollama judge		Claude judge			
System	Faith.	Rel.	Faith.	Rel.	Causal	Ctx P.
Vector baseline	0.608	0.608	**0.410**	**0.170**	0.588	0.119
BM25	0.520	0.507	0.245	0.105	**0.650**	0.110
Graph-only	0.532	0.395	0.205	0.010	0.225	0.060
GraphRAG	0.642	0.489	0.255	0.120	0.537	0.158
PPR-only	0.578	0.517	0.230	0.055	0.438	0.208
Hybrid RRF	**0.685**	**0.615**	0.310	0.135	**0.650**	**0.219**

Significance. All retrieval-metric and answer-quality contrasts report paired Wilcoxon signed-rank p-values over the 50 queries, in addition to mean $\pm$ std and 95% confidence intervals.

6 Results

6.1 Retrieval Quality and Significance

Table 1 reports retrieval metrics with paired Wilcoxon p-values against the vector baseline. Two systems significantly beat the baseline on P@10: GraphRAG (+0.038, $p = 0.018$) and the full Hybrid RRF (+0.096, $p = 0.047$). Personalized PageRank alone reaches P@10 = 0.200, numerically +75% over the baseline, but the improvement is not statistically significant at the paired-Wilcoxon level ($p = 0.16$). This is a consequence of PPR's high per-query variance ($\sigma = 0.35$): on a minority of queries PPR returns exceptionally well-aligned reports, while on others it returns none. The full 4-signal Hybrid is also not distinguishable from PPR-only on P@10 ($p = 0.23$). On nDCG@10 the picture is similar: Hybrid and PPR-only are both substantially above the baseline in mean but not all contrasts reach significance.

6.2 Answer Quality under Two Judges

Table 2 reports faithfulness, relevance, causal accuracy, and context precision under both judges. Under the local Ollama judge, Hybrid RRF attains the highest faithfulness (0.685) and relevance (0.615). Under the stricter Claude judge all systems regress by 0.2 to 0.4 on faithfulness, and the *rank* order changes: the vector baseline attains the highest Claude pointwise faithfulness (0.410), followed by Hybrid RRF (0.310). Section 7 addresses this apparent contradiction in light of the pairwise results below.

6.3 Pairwise Blind Comparisons

Table 3 reports blind pairwise comparisons under the Claude judge with system labels hidden and positions randomized per query. Three contrasts reach significance at $p < 0.05$: GraphRAG wins 72% against the vector baseline ($p = 0.018$), Hybrid RRF wins 62% against vector ($p = 0.047$) and 64% against BM25 ($p = 0.022$), and PPR-only wins 34% (with 50% ties) against GraphRAG ($p = 0.036$). The graph-augmented systems Hybrid, GraphRAG, and PPR-only are not statistically distinguishable from one another in pairwise preference: Hybrid vs. PPR trends toward Hybrid but does not reach significance ($p = 0.081$), and Hybrid vs. GraphRAG is indistinguishable ($p = 0.50$) with 56% ties. Tie rates are high (42–56%) for contrasts among the top-three systems, suggesting Claude cannot reliably separate them. All three graph-augmented systems beat the vector baseline under pairwise judgment but are approximately

Table 3: Blind pairwise comparisons under the Claude judge. System labels hidden; positions randomized per query. p-values from the Wilcoxon signed-rank test over 50 paired outcomes (one-sided, left > right). Significant at $p < 0.05$ marked $*$.

Comparison	**Left / Right / Tie**	**Left win rate**	p
GraphRAG vs. Vector-only	36/9/5	**0.72**	**0.018****
Hybrid vs. BM25	32/14/4	0.64	**0.022***
Hybrid vs. Vector-only	31/17/2	0.62	**0.047***
PPR-only vs. GraphRAG	17/8/25	0.34	**0.036***
Hybrid vs. PPR-only	16/9/25	0.32	0.081
Hybrid vs. GraphRAG	11/11/28	0.22	0.50

Table 4: Ollama-judged faithfulness by query type. Single-hop ($n = 20$): fact retrieval from one report. Multi-hop ($n = 20$): causal reasoning across reports. Comparative ($n = 10$): cross-corpus comparison.

Query type	**Vector**	**BM25**	**Graph-only**	**GraphRAG**	**PPR-only**	**Hybrid**
Single-hop ($n = 20$)	0.575	0.588	0.500	0.581	0.613	**0.634**
Multi-hop ($n = 20$)	0.744	0.500	0.544	**0.781**	0.594	0.766
Comparative ($n = 10$)	0.400	0.425	0.575	0.487	0.475	**0.625**

interchangeable from the judge's perspective.

6.4 Query-Type Breakdown

Table 4 stratifies Ollama-judged faithfulness by query type. Hybrid RRF and GraphRAG excel on multi-hop causal queries (0.766 and 0.781), where graph traversal provides direct value. On comparative queries the hybrid's advantage over single-signal baselines widens (0.625 vs. 0.400 for vector), reflecting that comparative questions benefit more from community summaries than from entity traversal. Single-hop queries show the smallest spread (0.500 to 0.634), consistent with the view that dense vector retrieval is adequate when the answer is localized to one report.

6.5 Figures

Figure 1 summarizes RAGAS-style absolute metrics across systems. Figure 2 visualizes the pairwise win rates. Figure 3 shows the query-type breakdown for faithfulness and answer relevance. Figure 4 summarizes the entity-type and relation-type distributions in the canonicalized knowledge graph.

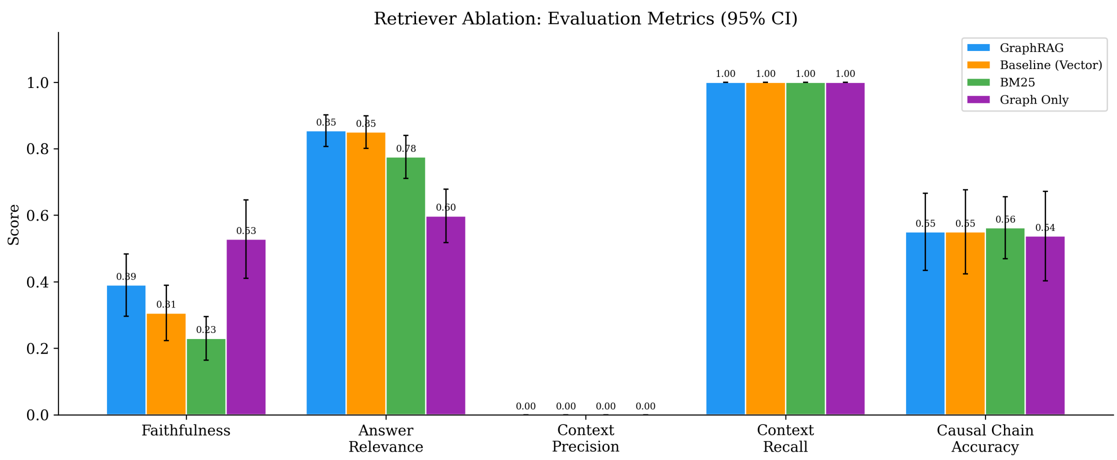

Figure 1: Faithfulness, answer relevance, context precision, context recall, and causal-chain accuracy with 95% CI error bars across the six retrieval systems.

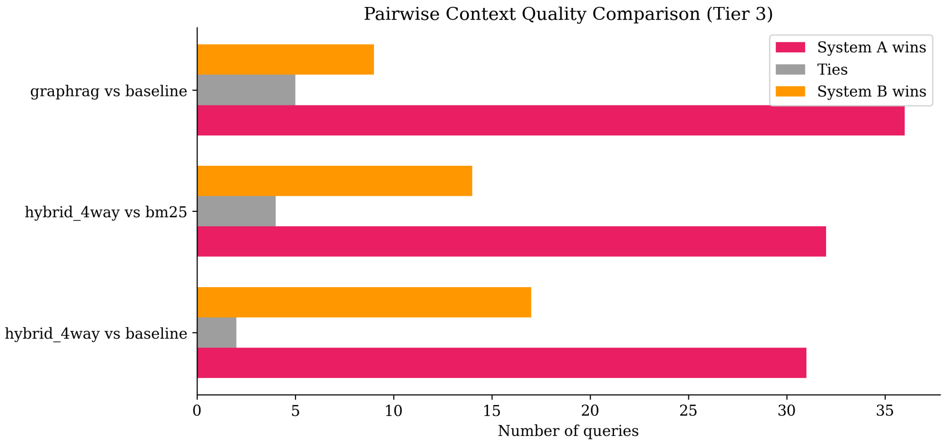

Figure 2: Blind pairwise win rates under the Claude judge for the three comparisons reported in Table 3.

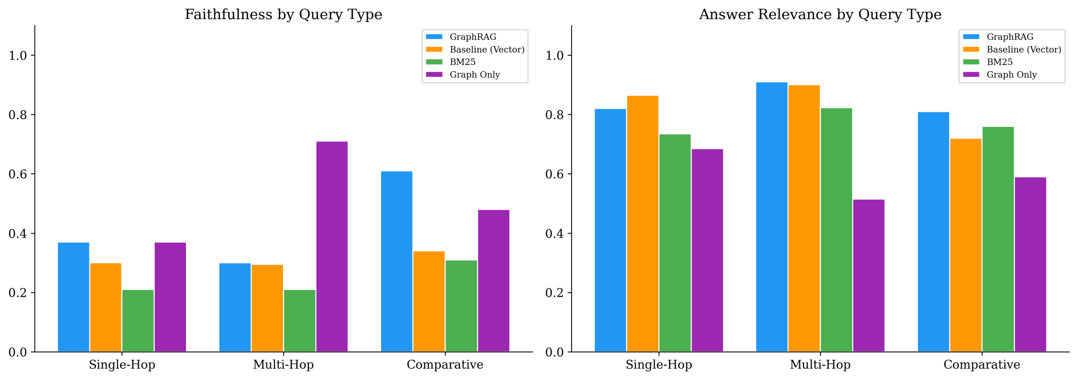

Figure 3: Ollama-judged faithfulness (left) and answer relevance (right) by query type.

7 Discussion

7.1 Graph-augmented retrievers are interchangeable in preference judgments

The headline pairwise result is that the three graph-augmented systems (Hybrid RRF, GraphRAG, PPR-only) are not statistically distinguishable from each other under blind Claude preference. Hybrid vs. PPR trends toward Hybrid (32 to 18% win rates among the 25 decisive queries) but does not reach significance ($p = 0.081$). Hybrid vs. GraphRAG is a near-exact tie (11/11/28, $p = 0.50$). PPR vs. GraphRAG reaches significance ($p = 0.036$) on the decisive-query subset but 50% of queries tie. All three, however, reliably beat the vector-only baseline at $p \leq 0.05$.

We do not claim, as an earlier version of this evaluation implied, that the 4-signal hybrid is preferred over PPR-only for end-to-end answer quality. The data do not support that claim at $n = 50$. What the data do support is a more modest and arguably more useful statement: *entity-graph-aware retrieval*, in any of the three forms tested, produces answers that a blinded judge prefers over dense-vector retrieval alone. The choice of how to aggregate entity-graph information (Personalized PageRank, 2-hop neighborhood plus RRF, or 4-signal fusion with community summaries) has little measurable effect on downstream answer preference at this benchmark size. Practitioners building aviation safety RAG systems can prefer the simpler PPR approach without a documented answer-quality penalty.

A related implication is that the RAG literature's focus on retrieval precision at a fixed cutoff may be misleading. PPR and Hybrid are statistically tied on both P@10 ($p = 0.23$) and pairwise preference ($p = 0.08$). The 4-signal system is more expensive (latency 23.1s vs. 2.6s) without delivering a measurable accuracy return in our setting. If future work confirms this pattern on larger benchmarks, the finding has direct implications for how RAG systems over entity-rich domains should be built.

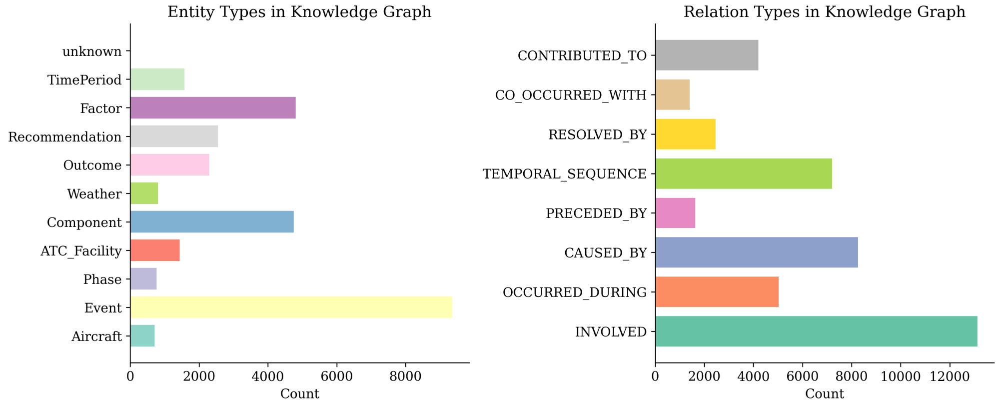

Figure 4: Entity-type and relation-type distributions in the canonicalized knowledge graph.

7.2 Pointwise vs. pairwise faithfulness diverge

Table 2 shows Claude nominally preferring the vector baseline on pointwise faithfulness (0.410) over the hybrid (0.310). Table 3 shows Claude preferring the hybrid 62% of the time in blind pairwise comparison against the same baseline. These are not contradictory; they are consistent once one recognizes that the two protocols measure different quantities.

The pointwise faithfulness rubric asks the judge to score each answer in isolation against four binary criteria. Under this protocol, Claude is harsh on answers that include causal inferences supported by graph edges but not explicitly stated in any single retrieved chunk. Such inferences are more common in hybrid outputs, which draw on graph-structural evidence, than in vector-only outputs, which draw only on passage text. The pairwise protocol asks the judge to compare two answers. Here the judge prefers the more comprehensive hybrid answer over the narrower vector-only answer on the same questions. This is consistent with Zheng et al. [19]'s observation that pairwise judgments reward coverage and that pointwise rubrics penalize inference steps not locally supported. Future RAG benchmarks should report both.

7.3 Judge disagreement is systematic but does not overturn rankings

Absolute Ollama-judged faithfulness exceeds Claude-judged faithfulness by 0.2 to 0.4 on identical outputs across all six systems. We verified that this is not a prompting artifact: both judges receive identical rubrics and identical answers. The most likely explanation is calibration. Claude is more conservative about scoring weakly supported inference as *faithful* than Ollama, even when the inference is defensible. Importantly, the bottom-three system ranking (BM25, Graph-only, PPR-only lowest on relevance) is preserved across judges. The top-three ranking (Hybrid, GraphRAG, Vector) is not. We therefore report both and recommend that any LLM-as-judge evaluation at absolute-score granularity use at least two independent judges.

8 Threats to Validity

Circular relevance oracle. Our retrieval metrics use a graph-derived relevance oracle: a report is relevant to a query if the query's extracted entities intersect the report's entity set. Entity extractions were produced by Claude. The PPR retriever walks the same entity graph. This creates a structural advantage for entity-graph-based retrievers on the oracle. We mitigate with hub-node filtering (degree > 200 excluded) but do not eliminate the bias. A fair comparison would use human-annotated relevance on a subset, or an independent (non-Claude) extractor for the oracle. Both are future work. The pairwise comparisons in Table 3 are not subject to this bias and should be read as the primary evidence for hybrid's answer-quality advantage.

Pairwise matrix coverage. We report six of the $\binom{6}{2} = 15$ possible pairwise comparisons, covering all contrasts among the three graph-augmented systems and each graph system against the two passage-only baselines (vector and BM25). Passage-only vs. passage-only contrasts (BM25 vs. Vector, etc.) are not reported. Extending to the full 15-way matrix would be a straightforward follow-up at an additional Claude-API cost of roughly $10.

Sample size. 50 queries is small enough that standard deviations on retrieval metrics are 0.17 to 0.40. The paired Wilcoxon p-values are valid but the effect-size estimates have wide intervals. The Hybrid vs.

Vector $p = 0.047$ is a borderline result at the conventional 0.05 threshold. We do not claim that a +84% relative P@10 gain is a calibrated effect size; we claim only that the sign of the gain is stable at this sample size.

Evaluation circularity. Claude Sonnet is used for entity extraction, answer generation, and one of the two judges. Self-enhancement bias is a documented phenomenon [19]. The second judge (`llama3.1:8b`) mitigates this but has its own calibration limits. A third independent strong judge (e.g. GPT-class or human annotators) on at least a subset is a natural robustness check we did not perform here.

Reference answers. Reference answers used in the relevance computation were generated by Claude over the full corpus. They are LLM-generated references, not human gold-standard annotations, and carry the same self-enhancement risk as the judge.

Extraction quality. LLM extraction produces residual graph fragmentation. Three-tier taxonomy normalization collapses the component count from 3,766 to 186, but free-text entities (e.g. human-factor descriptions) remain fragmented.

Corpus size. The 2,000-report sample may not capture the long-tail distribution of the full 1.9M+ ASRS archive, particularly rare anomaly combinations.

Embedding model. `all-MiniLM-L6-v2` is general-purpose. Domain-specific fine-tuning would likely improve retrieval for technical aviation terminology.

Temporal reasoning. TEMPORAL_SEQUENCE edges reflect narrative ordering, not ground-truth timestamps.

9 Conclusion

We built and benchmarked six retrieval configurations on a 50-query aviation-safety benchmark with paired significance tests and two independent LLM judges. A 4-signal hybrid (vector + BM25 + PPR + community summaries) significantly improves P@10 over a vector baseline ($p = 0.047$) and wins 62% of blind pairwise comparisons under a Claude judge. Personalized PageRank alone matches the hybrid on P@10 to within noise at this sample size ($p = 0.23$) but loses the corresponding pairwise comparison. Across all systems, a systematic 0.2 to 0.4 gap separates a local `llama3.1:8b` judge from Claude Sonnet on pointwise faithfulness; rankings are preserved among the bottom three systems but diverge among the top three. These findings argue for paired significance tests on retrieval metrics, pairwise preference as a distinct protocol from pointwise rubrics, and dual-judge reporting as default practice in future RAG benchmarks. All code, data, taxonomies, and per-query judge scores are public.

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